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Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, JUNE 2025

MCA Semester : 2

23MXAM OPEN SOURCE SYSTEMS

Time : 3 Hours **Maximum Marks :100**

INSTRUCTIONS:
1. Answer **ALL** questions. Each question carries 25 Marks.

1. a) Compare and contrast proprietary softwares with open source softwares. (3)

b) i) State and explain with example copy left licenses. (3)

ii) With an example explain the steps involved in bug life cycle in software testing. (7)

c) i) Why does cathedral model not much suitable like bazaar model in promoting OSS contribution and collective problem solving within open source communities? (12)

(OR)

ii) Elaborate on the need for open-source software licenses. Analyze the different licenses and its purpose with example software.

2. a) Write about the key features of R that make it a popular choice for data scientists. (3)

b) i) Write the proper R code to create a matrix that stores book details like bookID, bookTitle, UnitPrice, YearPublished as follows:

bookID	edition	Unit Price	YearPublished
1201	4	550	2023
1202	2	1220	2024
1203	1	3210	2025

Also determine proper R script to find the total cost of all books, and the books published after 2023 (7)

ii) Identify the output of the following R code and explain what the function lapply does here. data_list<- list(num = c(1, 2, 3), log = c(TRUE, FALSE), char = c("a", "b"))

lapply(data_list, function(x) x[1]) (3)

c) Describe in detail on how to handle CSV files in R. Assume you have a CSV file with student mark details like studID, studName, Cours1, Course2, Course3, Percentage to store the academic score in 3 different courses and overall percentage earned. Write proper (12)

i) read the csv file and store it as data frame.

ii) update any one course scores of a particular student.

iii) calculate the average score of each course .

iv) list the courses which has all scores greater than 60

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3 a) Why is NoSQL better choice for Big data analysis? (3)

b) i) Justify how a replicated Mongo DB could keep users intact by prioritizing availability and allowing eventual consistency in case of network partition. (3)

ii) How does two-Phase Commit protocol ensure transactional integrity and session consistency in distributed NoSQL databases? (7)

c) With proper example describe the aggregation and indexing methods in MongoDB. (12)

4 a) State the purpose of any two web frameworks on their use for developers. (3)

b) i) Describe the role of automated testing in open-source software development. (3)

ii) With a neat architecture diagram describe how Joomla offers content management (7)

c) How Gitworks as a distributed version control system? Analyze on the different storage areas Git uses to save the code before and after commit. (12)

/END/

FD/RL